Learning goals:

- Understand and apply common steps along an decoding pipeline for oscillatory responses;
- Understand common spatial patterns (CSP);
- Get familiar with the MNE toolbox for EEG analysis;
- Get familiar with sklearn for EEG decoding.

Question 1: Refresh you understanding

Refresh your understanding in the following concepts:

- Variance and covariance: Ben Lambert
- (generalized) eigenvalue decomposition: 3Blue1Brown, Ghojogh et al. 2023, Khan Academy
- Rayleigh quotients: Guanglian Chen

Question 2: Properties of CSP

(a) Write down the objective function of CSP.

(b) For each variable in the objective function, provide the dimensionality and the conceptual meaning.

(c) Does CSP have an analytical solution?

(d) What are the assumptions made by CSP?

(e) Is CSP a supervised or unsupervised method?

(f) Is CSP found in a classification or regression pipeline?

Question 3: Filters versus patterns

(a) How do spatial filters and patterns relate to the backward and forward problem?

(b) Does CSP obtain a spatial filter or pattern?

(c) Why can we not interprat a spatial filter?

(d) How do we obtain a spatial pattern from a spatial filter?

Question 4: Jupyter notebook CSP

Now put the theory into practice by going through the `CSP` Jupyter notebook.

Question 5: Submitting to Brightspace

For this assignment, please submit the following:

- The Jupyter notebook with your solutions
- A report (pdf extension) with the answers to the theoretical questions. In this report, add a line stating who contributed to what (please, adhere to the code of conduct). Additionally, please add a line on whether, where, and how you used GenAI for this assignment.

Question 6: (Optional) Test your understanding with real exam questions

(a) CSP

- 2020-2021 Part 5
- 2021-2022 resit Part 3
- 2022-2023 Part 4
- 2025-2026 Part 5